



Analytics Final Exam Study Guide

17 Questions (8 T/F Conceptual, 1 Graphical, 8 Computational), 120 Minutes, Lowest Scoring Dropped

8 questions on the material since the 2nd exam (GLMs, Categorical GLMs, Count GLMs).

Final Exam Question Distribution:

SOA Topic 1: Basics of Statistical Learning (22%) (4 Qs)

- a) Define terms used to classify the types of modeling problems and methods, including supervised versus unsupervised learning and regression versus classification. (1 Q)
- b) Compare the common methods of assessing model accuracy. (1 Q)
- c) Understand how the bias-variance tradeoff impacts the selection of statistical learning methods. (1 Q)
- d) Understand resampling methods used for model validation, including: Training set vs. test set approach, k-fold cross-validation, Leave-one-out cross-validation (1 Q)

SOA Topic 2: Linear Models (78%) (13 Qs)

- a) Compare model assumptions for ordinary least squares and generalized linear models. (1 Q)
- b) Identify the members of the exponential family of distributions and corresponding link functions. (1 Q)
- c) Apply the business context of a problem to interpret parameters. (1 Q)
- d) Interpret diagnostic tests of model fit and assumption checking, using
 - Graphical methods (1 Q)
 - Quantitative methods (1 Q)
- e) Select an appropriate model, considering
 - Distributions and link functions (1 Q)
 - Variable transformations and interactions (1 Q)
 - t and F tests (1 Q)
 - AIC and BIC (1 Q)
 - Likelihood ratio test (1 Q)
- f) Calculate and interpret predicted values, and confidence and prediction intervals. (2 Q)
- g) Understand how approaches may differ compared to using an ordinary least squares model, including
 - Regularized regression (lasso, ridge regression) (0 Q)
 - K-nearest neighbors (1 Q)



Formulas to Remember:

- Bayes Error Rate, Distance Formula for KNN
- Residual: $e_i = y_i - \hat{y}_i$
- $SST = SSE + SSReg$ and each's definition
- $s_y^2 = \frac{SST}{n-1}$
- $r = \frac{Cov(X,Y)}{s_x s_y}$ and using your calculator
- $R^2 = \frac{SSReg}{SST}$
- $se(b_0) = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- $t = \frac{\text{Estimator} - H_0 \text{ value}}{se(\text{Estimator})}$
- $se(\hat{y}_{pred}) = s \sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- $R_a^2 = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - (1 - R^2) \left(\frac{n-1}{n-(p+1)} \right)$
- Whole $F = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{\frac{R^2}{1-R^2}}{\frac{p}{n-(p+1)}}$
- Nested $F = \frac{\frac{(SSE_0 - SSE_{full})}{q}}{\frac{SSE_{full}}{n-(p+1)}} = \frac{\frac{R_{full}^2 - R_{reduced}^2}{q}}{\frac{1 - R_{full}^2}{n-(p+1)}}$
- $se(b_j) = s \sqrt{(j+1)st \text{ diagonal of } (X'X)^{-1}}$
- Confidence Interval: $b_j \pm t_{critical} * se(b_j)$
- $r_{partial} = \frac{t_i}{\sqrt{t_i^2 + n-(p+1)}}$
- Breusch-Pagan: Reject H_0 if $\frac{SSReg_{aux}}{2} > \chi_{\alpha,p}^2$
- Standardized Residual $e_{std} = \frac{e_i}{se(e_i)} = \frac{y_i - \hat{y}_i}{s \sqrt{1 - h_i}}$
- SLR Leverage $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2}$
- Cook's Distance $D_i = \left(\frac{e_i}{s \sqrt{1 - h_{ii}}} \right)^2 \frac{h_{ii}}{(p+1)(1 - h_{ii})}$
- $VIF_j = \frac{1}{1 - R_j^2}$
- AIC/Mallow's $C_p = \frac{1}{n} (SSE_p + 2ps_{full}^2)$
- $BIC = \frac{1}{n} (SSE_p + \ln(n) ps_{full}^2)$



- $CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$
- LOOCV: $CV = \frac{1}{n} \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$
- GLM: $\mathbf{x}'_i \boldsymbol{\beta} = g(\mu_i)$, $E[y] = b'(\theta)$, $Var(y) = \phi b''(\theta) = \phi v(\mu_i)$
- Linear Exponential Decomposition: $f(y; \theta, \phi) = e^{\left[\frac{y\theta - b(\theta)}{\phi} + S(y, \phi) \right]}$
- Canonical links, Mean-Variance Relations
- Tweedie: $v(\mu) = \mu^p$ for $1 < p < 2$
- Deviance: $D = 2(l_{SAT} - l_{\hat{\theta}})$
- $e_{i, Pearson} = \frac{y_i - \hat{\mu}_i}{\sqrt{Var(y_i)}}$
- $d_{i, Deviance} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{D}$
- $R^2_{maxscaled} = \frac{1 - e^{-\frac{2(l_0 - l)}{n}}}{1 - e^{-\frac{2l_0}{n}}}$ and $R^2_{pseudo} = \frac{l_{\hat{\theta}} - l_0}{l_{SAT} - l_0}$
- $LRT = 2(l_a - l_0) = D_0 - D_a \sim \chi^2_q$
- Logit: $g(\mu) = \ln \frac{\mu}{1-\mu}$, $\hat{\mu}_i = \frac{1}{1+e^{-z}}$ where $z = b_0 + b_1 x_1 + \dots + b_p x_p$
- Probit: $g(\mu) = N^{-1}(\mu)$, $\hat{\mu}_i = N(z)$ where $z = b_0 + b_1 x_1 + \dots + b_p x_p$
- Complementary Log-Log: $g(\mu) = \ln(-\ln(1-\mu))$
- Odds = $\frac{p}{1-p}$ and $p = \frac{\text{Odds}}{1+\text{Odds}}$
- Bernoulli $l(\boldsymbol{\beta}) = \sum_i [y_i \ln \mu_i + (1-y_i) \ln(1-\mu_i)]$
- Bernoulli $D(\hat{\boldsymbol{\mu}}) = 2 \sum_i \left[y_i \ln \frac{y_i}{\hat{\mu}_i} + (1-y_i) \ln \frac{1-y_i}{1-\hat{\mu}_i} \right]$
- Bernoulli Pearson $e_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1-\hat{\mu}_i)}}$
- Logit Proportional Odds: $\text{logit}(F_Y(j)) = \ln \frac{F_Y(j)}{1-F_Y(j)} = \ln \frac{\pi_1 + \pi_2 + \dots + \pi_j}{\pi_{j+1} + \dots + \pi_c} = \alpha_j + \mathbf{x}'_i \boldsymbol{\beta}$
- Log-link Offset: $\ln \mu_i = \ln E_i + \mathbf{x}'_i \boldsymbol{\beta}$
- Poisson $l(\mathbf{b}) = \sum_{i=1}^n [y_i \ln \mu_i - \mu_i - \ln(y_i!)]$
- Score Equations: $\sum_{i=1}^n e_i = 0$ and $\sum_{i=1}^n x_i e_i = 0$
- Poisson $D = 2 \sum_{i=1}^n \left[y_i \ln \left(\frac{y_i}{\hat{\mu}_i} \right) - (y_i - \hat{\mu}_i) \right]$
- Poisson Pearson $e_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$
- Poisson Overdispersion: $\hat{\phi} = \frac{1}{n-(p+1)} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}$
- Zero-Inflated: $P(y_i = j) = \begin{cases} \pi_i + (1-\pi_i)g_i(0), & j = 0 \\ (1-\pi_i)g_i(j), & j = 1, 2, \dots \end{cases}$
 $E[y_i] = (1-\pi_i)\mu_i$, $Var(y_i) = (1-\pi_i)\mu_i + \pi_i(1-\pi_i)\mu_i^2$
- Hurdle: $P(y_i = j) = \begin{cases} \pi_i, & j = 0 \\ k_i g_i(j), & j = 1, 2, \dots \end{cases}$, $k_i = \frac{1-\pi_i}{1-g_i(0)}$, $E[y_i] = k_i \mu_i$, $Var(y_i) = k_i \mu_i + k_i(1-k_i)\mu_i^2$